G093_User Manual

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Introduction

In this User Manual, we're going to go over some of the details of this project, such as detailed information on how to play, what are the requirements for running this project and the features of this project. We're also going to include a troubleshooting tab, as well as a FAQ tab.

Program Overview

This program was designed to be a game about a railway management system. It includes the possibility to create cities, buy stations and upgrade such stations, giving them bonuses to enhance their use during the gameplay. We can also buy locomotives, which are used by trains to deliver cargo produced by industries and cities. These actions produce money, which makes the player's budget increase. This project has a very innovative part to it, which defines the tech limitations simulating different timestamps on human history.

System Requirements

Minimum System Requirements (for IntelliJ IDEA):

- OS: Windows 10/11, macOs 12+, Linux (various distributions allowed).
- Ram: 2GB Minimum, 8GB Recommended.
- Disk Space: Around 3.5GB + extra value for caches and the files themselves.
- JDK: Recommended to use the one bundled with IntelliJ IDEA.

Other Requirements:

- JavaFX (add the library to the project).
- Apache Commons Exec 1.4.0 (download the .jar).
- Graphviz-Java 0.18.1 (needs to be in the PATH) (also download the .jar).
- SLF4J 2.1.0-alpha1 (download the .jar).
- Gnuplot must be installed (needs to be in the PATH).

Features

The user can login into 2 roles: Player (who plays the game) and Editor (who prepares the game for the Player). The Editor is responsible to create the maps, place cities and their respective house blocks into the map, and create a scenario, which will be associated to one and only one map. Players can place different types of stations on a select map and scenario, as well as upgrade them. Positioned stations can be upgraded and connected by railway lines. After the cargo of a city/industry finishes the producing state, players can choose a locomotive and retrieve the cargo, to increase their budget and allow them to progress into the game.

Main Menu

The first menu the app presents is the general menu. Here, the user can login into an account (which can be an account from an Editor or from a Player), create a new account or even see the development team for this project.

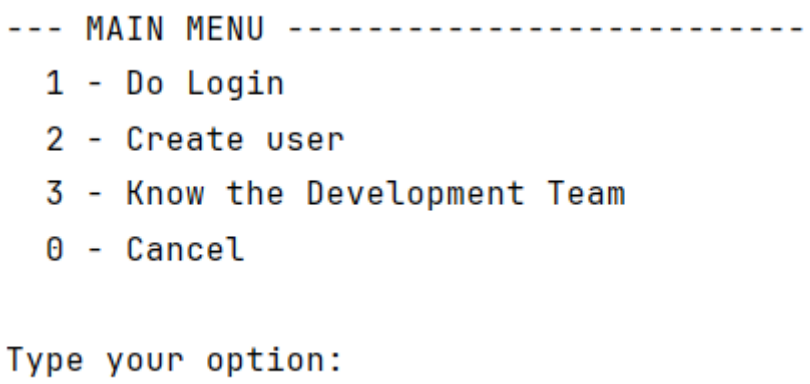


Figure 1 - Main Menu

After creating the account, the user can play. But beforehand, someone needs to create a map and scenario in which the player will play. That's the job of the editor.

The Editor

When the editor logs into his account, he is presented with a menu that showcases what he can do.

```
--- ADMIN MENU --------------------------
  1 - Create Map
  2 - Select Map
  0 - Cancel

Type your option:
```

Figure 2 - The Editor's Menu

1. Create Map

Create Map allows the Editor to define the width and height of a map. This data is important to the player, for him to understand the size of the map and being able to place his stations inside the map.

```
--- Create Map ------------------------
Enter Map Name: example
Enter Map Width: 100
Enter Map Height: 100
Enter Map Scale (km): 1

Map successfully created!
Do you want to save the map as a file? (y/n)
y
Map saved successfully!
```

Figure 3 - Example of Map Creation

2. Select Map

If the Editor selects this option, he will be asked to choose a map that belongs to the list of all maps that were created. Upon selection, a new menu will be presented to the Editor, with its main purpose being to help the Editor change the map, doing so by creating a Scenario or adding a City for instance. This is the menu presented:

```
--- ADMIN MENU --------------------------
  1 - Create City
  2 - Create Scenario
  0 - Cancel

Type your option:
```

Figure 4 - The Map Editor

2.1 Create City

Create City allows the Editor to place a city inside a certain map. He can choose the City's center coordinates, but he must respect the map size (can't place it outside the map) or in the same position as another city. Cities are also made of House Blocks, which are responsible for producing cargo. House Blocks can be placed manually or automatically. Manually, the editor chooses the X and Y coordinates himself, however in Automatic, the program runs a normal distribution to determine their positions.

```
--- Create City ------------------------
Enter City Name: example
Enter Center X: 22
Enter Center Y: 22
City created: example on map exampleMap
Choose block mode (M=Manual, A=Auto): A
Number of blocks: 3
Standard Deviation : 1
Blocks assigned automatically.
Generated block coordinates:
Block 1: (22, 23)
Block 2: (23, 23)
Block 3: (21, 22)
City created with success.
```

Figure 5 - Example of City Creation, using Automatic House Blocks

2.2 Create Scenario

Create Scenario allows the Editor to create a scenario for a specific map. Scenarios basically define the rules of the game, whether it being by specifying the types of locomotives the player can use, the start and finishing year of the simulation, and the historical restrictions (for example, if there is a school program happening, there will be an increase in production regarding cities). Not only that, but the menu also gives the ability for the editor to add multipliers, like the Production Rate and the Profit Multiplier. All these aspects matter, because each one of them impacts gameplay.

```
--- CREATE SCENARIO ------------------------

--- SCENARIO NAME --------------------------
Note: Scenario Name must be unique and contain only alphanumeric characters and spaces.
Enter Scenario Name: example

--- STARTING YEAR --------------------------
Note: Starting Year must be equal or higher than 1804 and lower than 2025.
Enter Starting Year: 1890

--- FINAL YEAR -----------------------------
Note: Final Year must be higher than the Starting Year and equal or lower than 2025.
Enter Final Year: 2000

--- PORT BEHAVIOR --------------------------
Note: Chosen behavior must be included in the following list:
1 - Port1 (All Cargo available to import and export)
2 - Port2 (Only PEOPLE and MAIL are allowed)
Enter Port Behavior: 1

--- AVAILABLE LOCOMOTIVE TYPES -------------------------
Note: Available Locomotive Types must be either 'steam', 'diesel', and/or 'electric'.
Enter Available Locomotive Types: steam,diesel

--- HISTORICAL EVENTS ----------------------
Note: Historical Events must be either 'none', 'vaccination campaign', and/or 'school program'.
Enter Historical Events: none
```

Figure 6 - Partial Creation of a Scenario

The Player

The player, as soon as he logs in, is presented with the Main Menu. In this menu, the player can either select the option to play the game or leave the game.

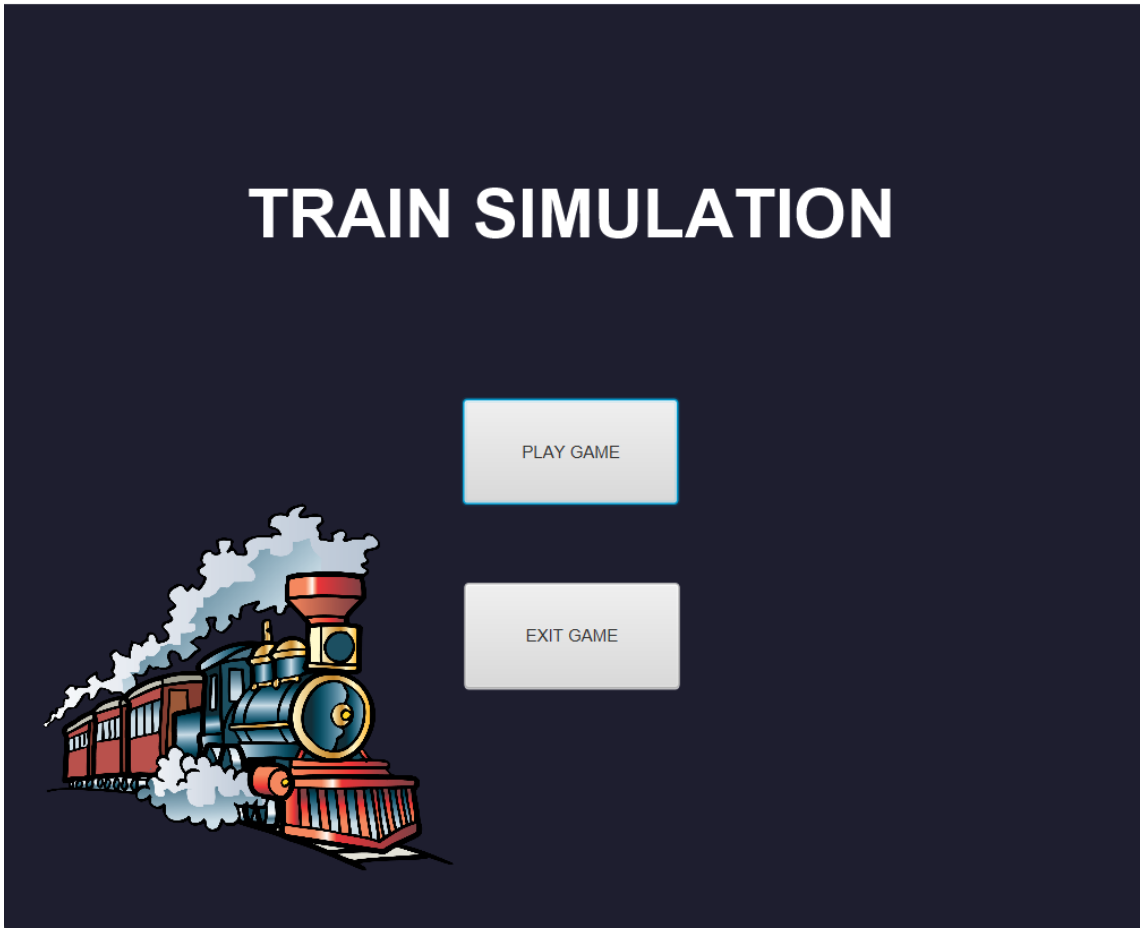


Figure 7 - The Player's Menu

If the player selects the option to play the game, a new menu will appear. This menu will ask the player to choose in which scenario he wants to play the simulation in. When selecting a scenario, its details are shown, including the Time Period, the restrictions and the multipliers.

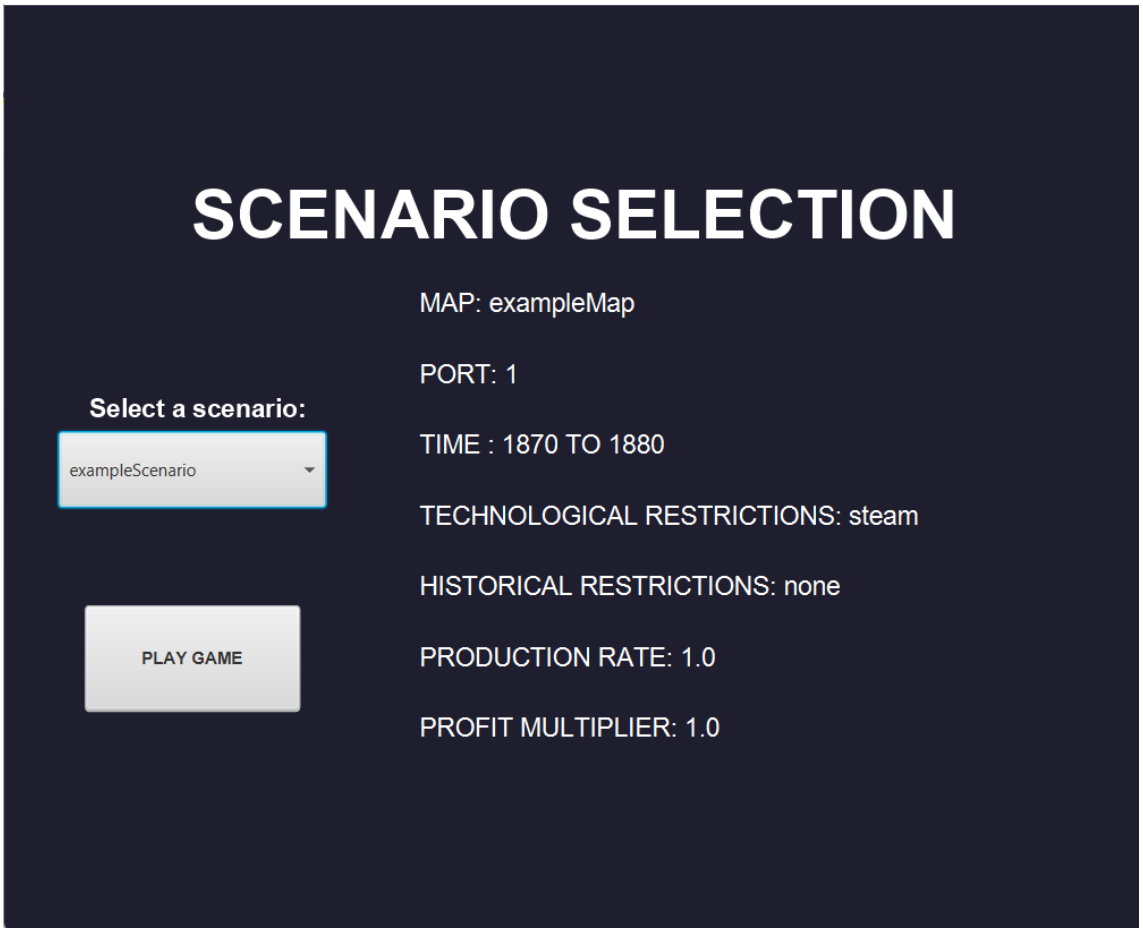


Figure 8 - Scenario Selection Menu

After the player makes his choice, the map will appear, as well as a new set of options, which will be used to play the game. As soon as the game starts, the time will start to count. The years will go by, and with it (if there are available stations) the cities and industries will start to produce cargo. This is where the strategic part of the gameplay kicks in. If he stays on the menu, the years will go on. The current year determines what locomotives he can buy and what type of buildings he can use. When the simulator reaches the final year for that specific scenario, the game finishes, sending him back to the main menu. However, it is important to notice that when the player is interacting with the menu itself, the cargo will stop to be produced, and the years will pause. Keep in mind that cargo demand will be updated every year, according to the deliveries of a certain type of cargo. After every year passes, the player must pay his yearly expenses. That includes the Locomotive Maintenance (according to what locomotive the player owns), the Track Maintenance (5€ for each piece of track), and the Fuel Cost (varies according to the Trains that are doing routes, and according to the distance they travel). If the player fails to pay for his yearly expenses, the game is over, and he needs to restart the simulation and rethink his strategy. This is the game menu he is presented with:



Figure 9 - The Game's Menu

1. Build a Station

The player can build three main types of station: depots, stations and terminals, each one having different prices and different capacities to storage cargo. Not only that, but each type of station also occupies a different area on the map, with Depots occupying a 3x3 area, stations a 4x4 area and terminals 5x5 area. The name of the station will be automatically generated, based on the nearest city and the type of station it is. After buying one, the map will update itself, showing the location of the newly bought station. There is no overbuilding allowed, which means there can't be a station in the same location as another station, city or an industry.

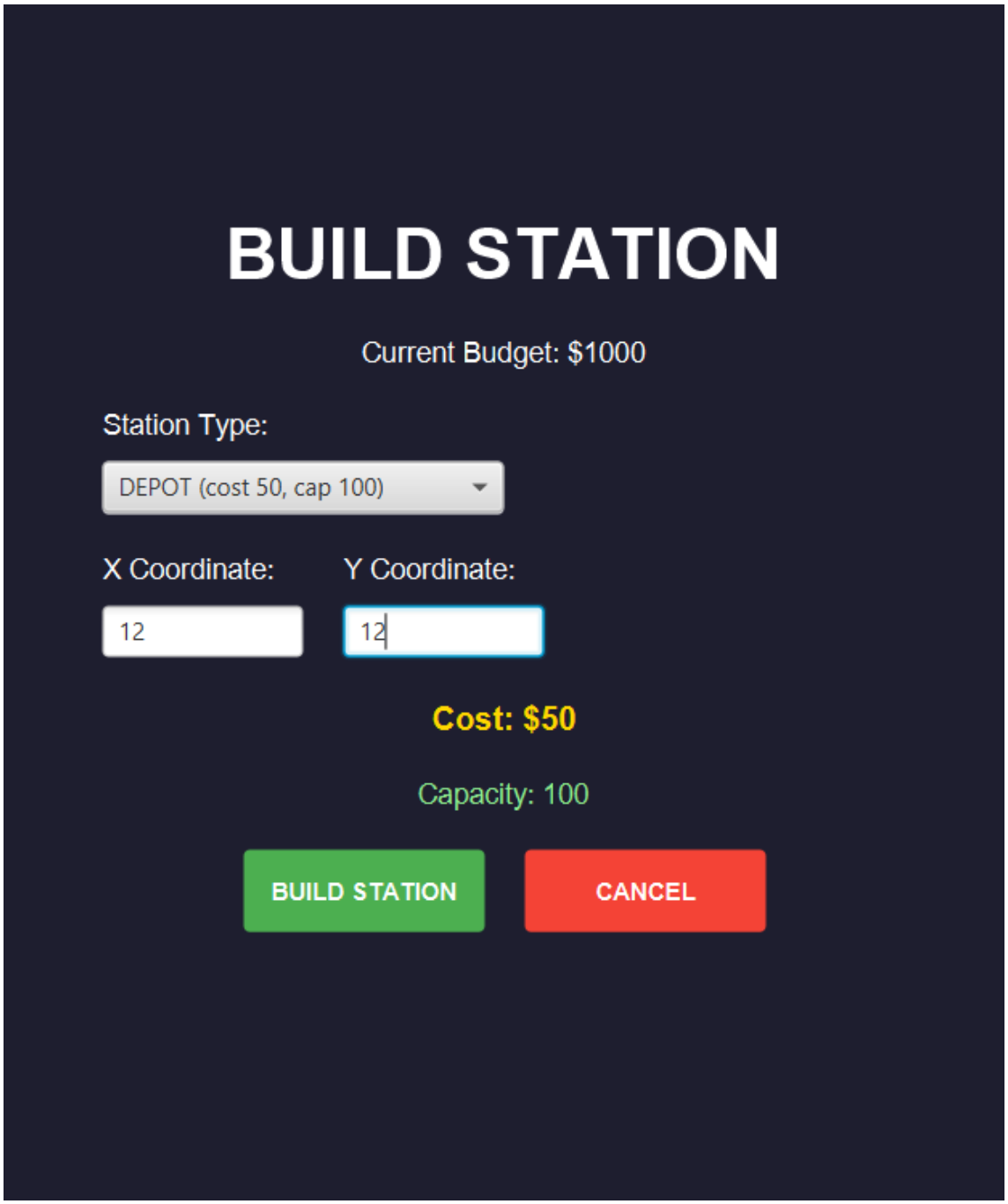


Figure 10 - Interface to Build a Station

2. Upgrade a Station (Type)

If the player has a Depot, however he wants to have a Station at the same location, he can do it. Upgrade Station (Type) allows the player to choose one of the stations he owns on that map and change its type. Depots can transform into Stations and Terminals, and Stations can become Terminals, but Terminals are not allowed to be upgraded in this part, since they are the best station that can be acquired.

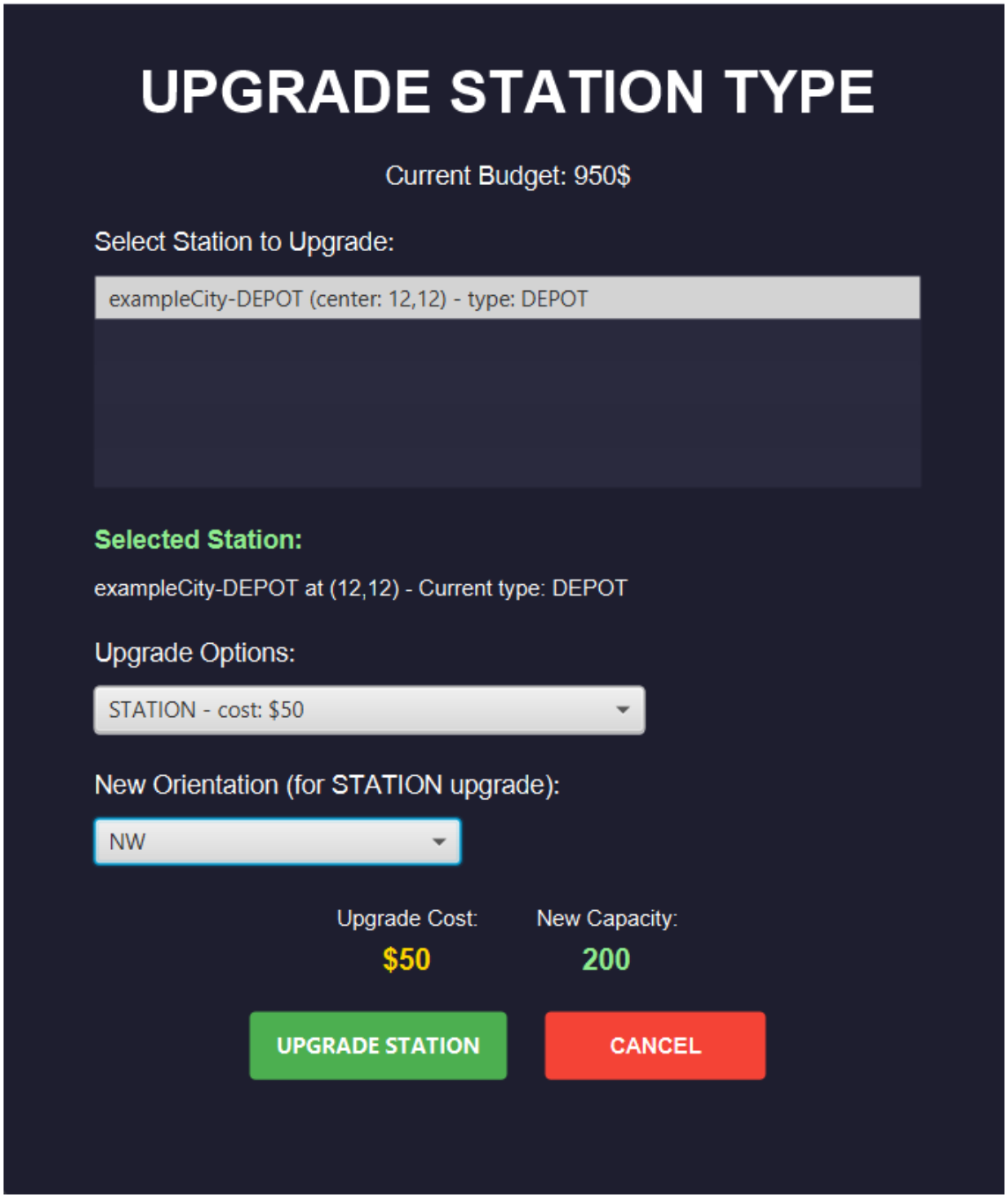


Figure 11 - Interface to Upgrade a Station's Type

3. Upgrade the Station (Building)

With this option, the player can choose one of the stations he owns on that map and upgrade it with a building. Buildings can grant a variety of bonuses to that station, allowing it to become more profitable. There are buildings that only become available at a certain year (for example the telephone, from when he starts to exist, the telegraph ceases to exist), or mutually exclusive (for instance the small and large café, when one is built, if a station has the other one, he will be discarded, to place the new one). Each station can only have one building of each type.

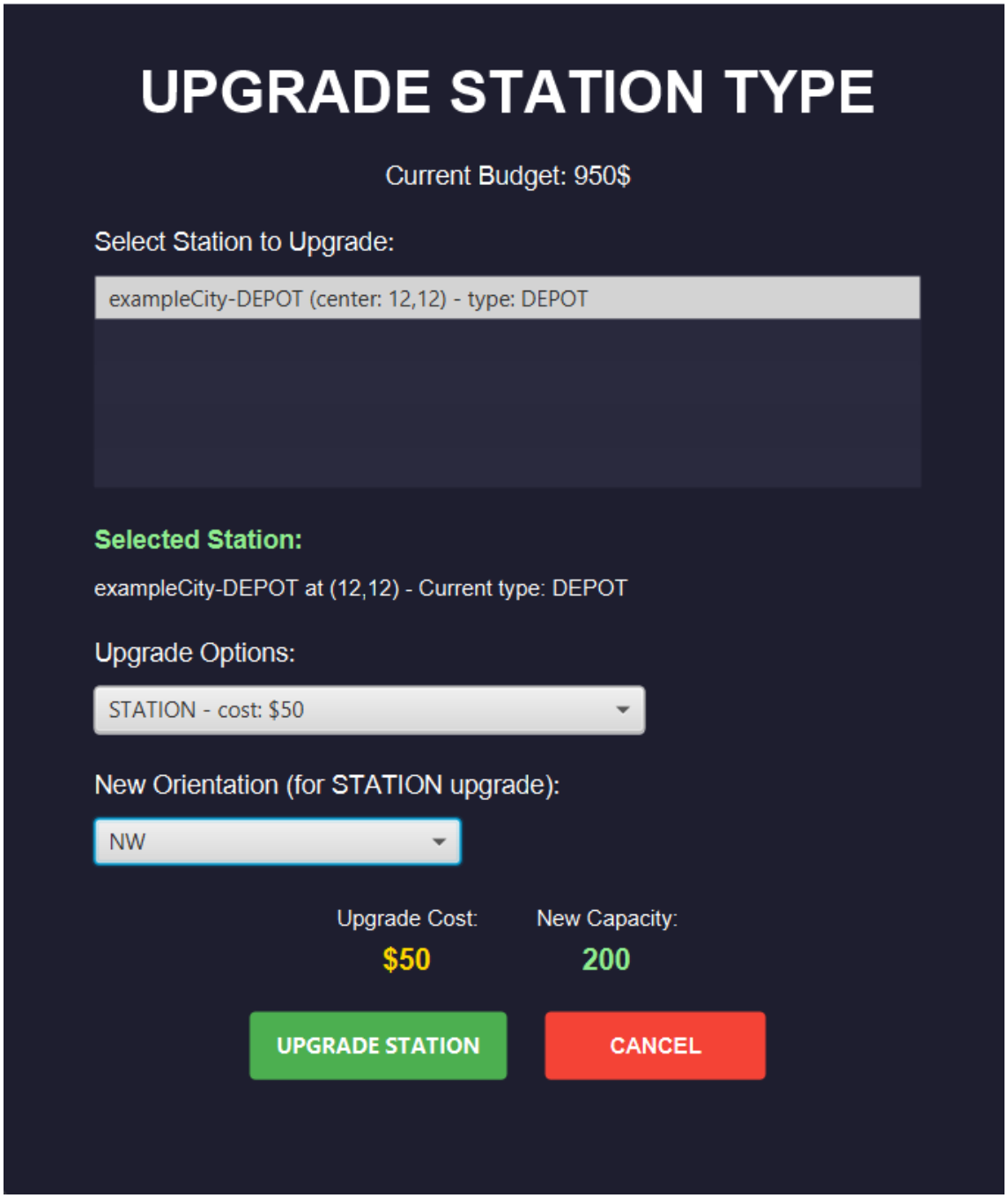


Figure 12 - Interface for Upgrading a Station with a Building

4. List Stations

The player can list all the stations he currently owns in a certain scenario. When the player chooses one, its details appear on a box, including the buildings that station has.

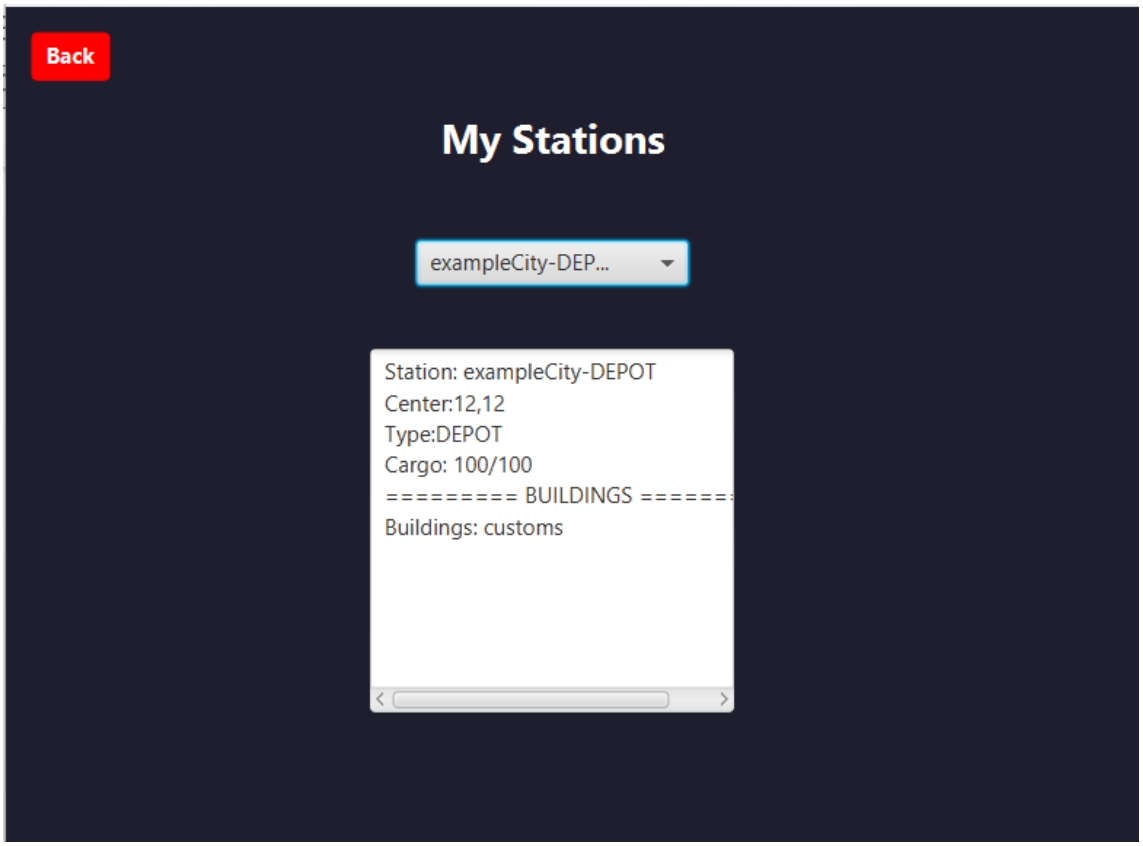


Figure 13 - Interface for Listing Stations

5. Buy a Locomotive

The main objective of the game is to earn more and more money, but to do that, after placing stations and letting them fill themselves with cargo, it's time to buy a Locomotive. In our project, players can pick from a variety of different locomotives, counting 65 in total. Every one of them has different top speeds, different types (steam, diesel or electric), different maintenance and fuel costs.

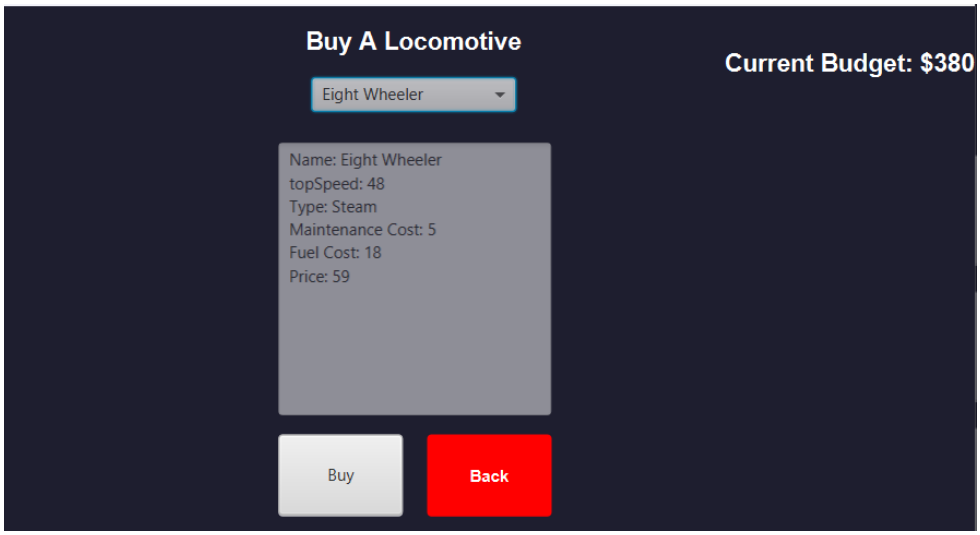


Figure 14 - Interface for Buying a Locomotive

6. List my locomotives

When the player starts to own a lot of locomotives, it can become hard to track the ones he owns and the ones he doesn't. Using this option, the software will display all the locomotives that the player owns. Not only that, the details of the locomotives are also displayed for the player to be able to see.

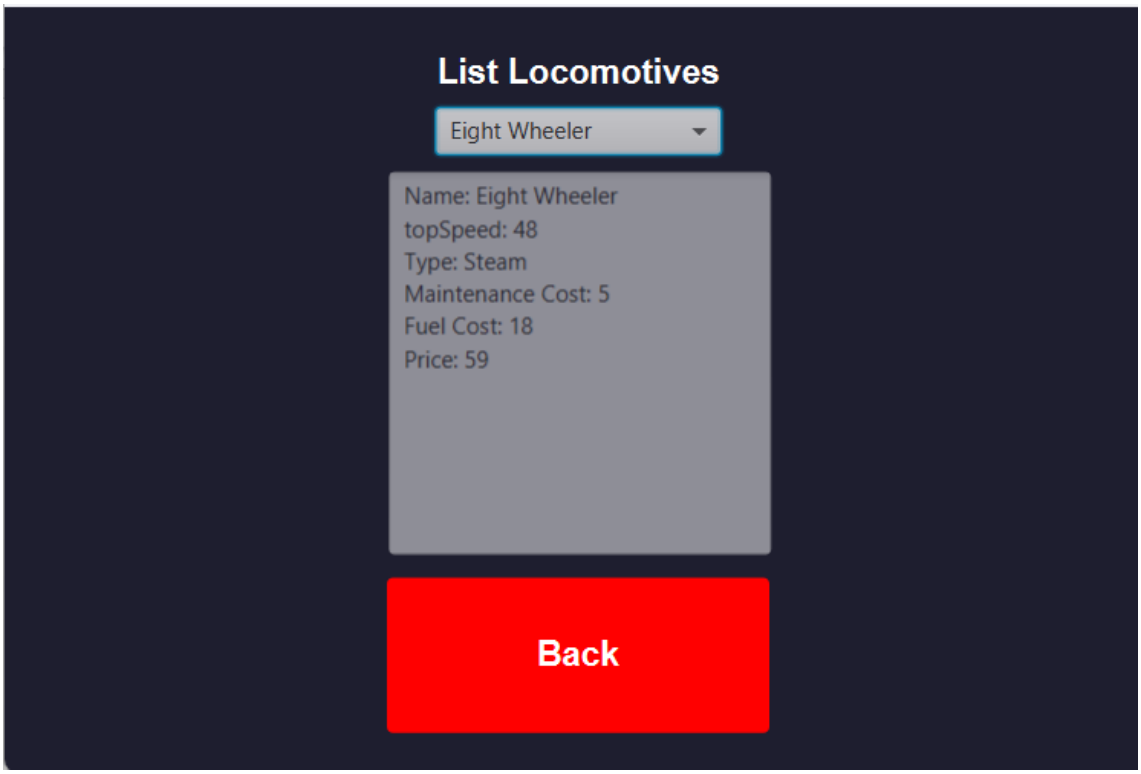


Figure 15 - Interface for Listing Locomotives

7. Associate a train with a route

This is where the player can increase his budget. When the player wants to, and there is cargo available to pick from at least 2 stations, the player can create a route. The route can be formed by only 2 stations, or by more. If there are more than 2 stations, the player can choose the starting and finish point of the route and decide whether there will be other stops in the middle. Beforehand, the player must choose a train to perform a route. Each train has a certain number of carriages of each cargo type, which determines the amount of cargo the train can pick up. For each point-of-route, the player can also choose a cargo pick up mode (FULL, HALF or AVAILABLE), which determines the behavior of said route.

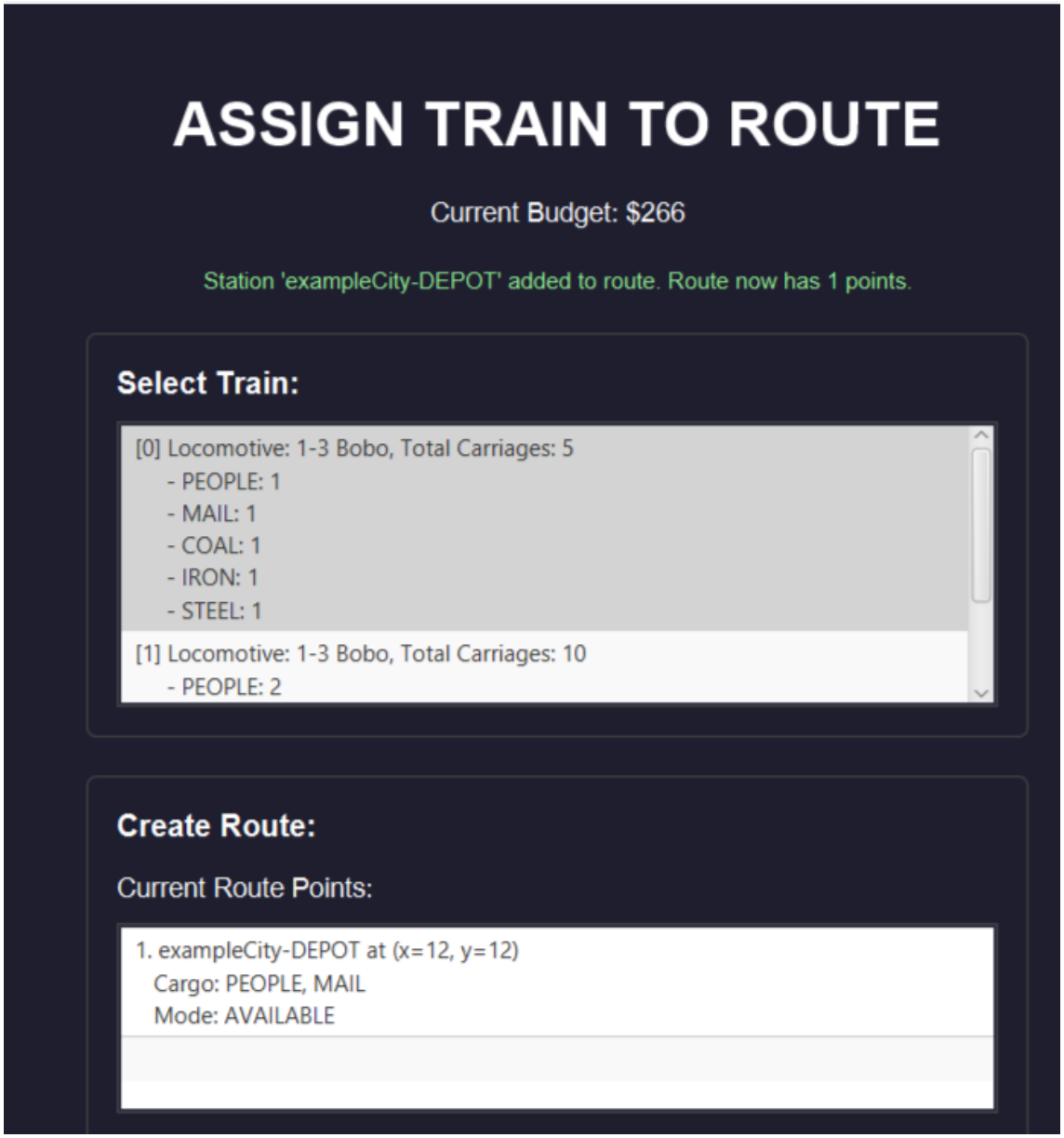


Figure 16 - Train Selection and Route Visualization

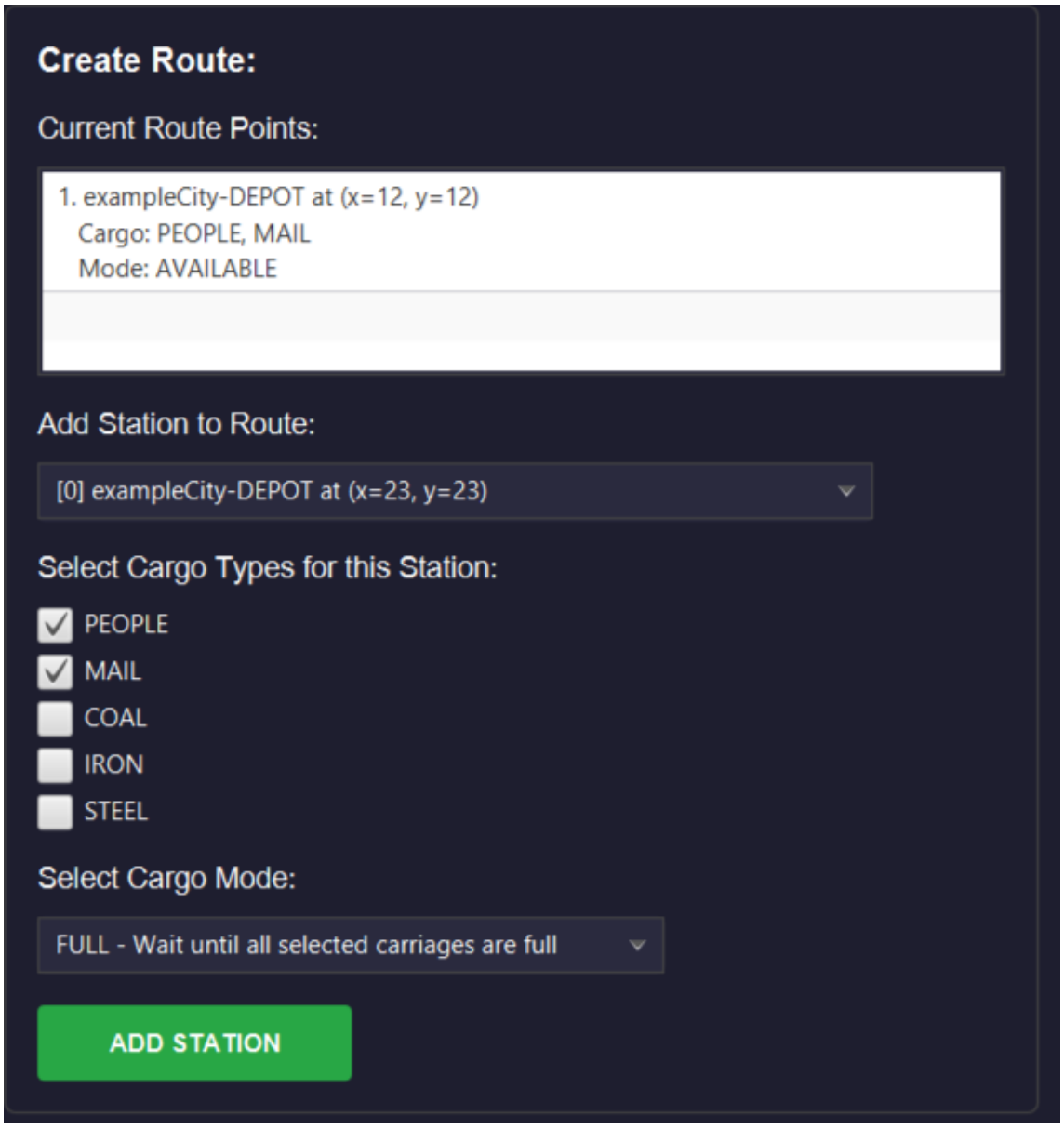


Figure 17 - Adding a New Station to the Route

8. Pause the game

The years will go by, but the player can change that. By using this function, the game will stop counting years and produce cargo. It's a practical functionality, since the player can stay away from the computer, but the years won't go by, and right after he comes back, he can press the option to "Resume Game" and pick up from where he left.

Troubleshooting & Helpdesk

- Missing the "*railway_connectivity* "folder
    - Issue: The game fails to start.
    - Solution: Ensure the folder is correctly placed at the root of the project.
- JavaFX not found
    - Issue: GUI doesn't render, or the project won't compile.
    - Solution: Add JavaFX library to your project manually.
- Graphviz/Gnuplot not recognized
    - Issue: Graphic visualization doesn't work.
    - Solution: Make sure graphviz/gnuplot is installed and the path to its bin folder is added to the System Path.

FAQ

Player Questions

- Q: What happens if I run out of budget?
- A: You won't be able to pay the yearly expenses, which means the game will be over. Review your expenses and station placement in order to improve!

- Q: How can I unlock better locomotives?
- A: New locomotives are unlocked as time progresses within the scenario.

- Q: Can I pause the simulation and return later?
- A: Yes you can, simply use the "Pause Game" option, which will stop time progression and cargo generation.

Editor Questions

- Q: Why can't I place a city?
- A: There are certain validations the game does before allowing such action. Make sure the city is within map boundaries, and you are not trying to build it above other structures.

- Q: How many cities are allowed to be on a map?
- A: There's no strict limit, but too many cities may slow down the performance of the simulation, especially if we are talking about a large map, with a simulation running on a lower-end system.

- Q: What is the difference between a Map and a Scenario?
- A: The map defines the layout and the size of the game world. A Scenario is the set of rules that will be applied to that map.